Cybersecurity Testbeds for Industrial Internet of Things

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Abstract

Cyber threats and attacks targeting industrial IoT require a flexible but scalable tool that allows the industry and research community to safeguard, improve security posture, and address cybersecurity compliance. The nature of today's attacks landscape respects no boundaries. Hence, the importance of cybersecurity testbeds is to facilitate security testing, vulnerability assessment, threat detection, education, and research. This paper presents a review of cybersecurity testbeds, Industrial IoT security goals, and challenges. The paper demonstrates the importance of cybersecurity testbeds in the field of cybersecurity, especially virtual cybersecurity testbeds. It was observed that the rapid development of the software industry to simulate various industrial components will continue to influence the application of virtual cybersecurity testbeds. Likewise, the quick adaptability of virtual cybersecurity testbeds would help cybersecurity practitioners and researchers to assess and mitigate risks to industrial systems while ensuring cyber preparedness and readiness. This paper provides insight and guidelines that may be used during the planning and designing of virtual cybersecurity testbeds.

Cybersecurity Testbeds for Industrial Internet of Things

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Abstract

Cyber threats and attacks targeting industrial IoT require a flexible but scalable tool that allows the industry and research community to safeguard, improve security posture, and address cybersecurity compliance. The nature of today's attacks landscape respects no boundaries. Hence, the importance of cybersecurity testbeds is to facilitate security testing, vulnerability assessment, threat detection, education, and research. This paper presents a review of cybersecurity testbeds, Industrial IoT security goals, and challenges. The paper demonstrates the importance of cybersecurity testbeds in the field of cybersecurity, especially virtual cybersecurity testbeds. It was observed that the rapid development of the software industry to simulate various industrial components will continue to influence the application of virtual cybersecurity testbeds. Likewise, the quick adaptability of virtual cybersecurity testbeds would help cybersecurity practitioners and researchers to assess and mitigate risks to industrial systems while ensuring cyber preparedness and readiness. This paper provides insight and guidelines that may be used during the planning and designing of virtual cybersecurity testbeds.

Keywords: Cybersecurity Testbeds, Industrial Internet of Things (IIoT), Cyber-Physical Systems Security,Industrial IoT Security, Industry 4.0 Cybersecurity, Operational Technology (OT) Security

1 Introduction

Assets categorized as critical infrastructure (CI) in today's environment consist of integrated systems and functions vital to the modern society's sustainability that, when interrupted, would affect the society's structure and economy [1]. Definitions of critical infrastructure by countries across the world vary. However, what is common among these definitions is that CI underpins the functioning of today's interconnected society and is integral for the human species' survival and safety [1]. For instance, critical infrastructure, as defined by The United States, consists of physical and cyber systems and assets. Unauthorized changes or alterations of such assets would negatively impact the people, structure, and economic activities of society. The impact of critical infrastructure when exploited has made it the target for cybercriminals. This has led to an increase in complex attacks demonstrating the professionalism and sophistication of the cyber-actors [1].

Target components for exploitation when it comes to critical infrastructure are the Industrial Control System (ICS), Supervisory Control and Data Acquisition (SCADA), Distributed Control System (DCS), and Operational Technologies (OT) [1]. However, integrating operational technologies and information technologies has further increased vulnerabilities in industrial IoT systems, hence the importance of assessing and mitigating attacks on industrial IoT systems.

Industrial IoT faces diverse security challenges such as unauthorized access, privacy, data theft, and safety issues [2],[3]. Addressing these issues has led to the use of cybersecurity testbeds. Cybersecurity testbeds have allowed the possibility of assessing and conducting security testing. This tool, however, has been deployed by industry professionals and researchers, but its relevance in the evergrowing attack landscape remains.

To understand the relevance of cybersecurity testbeds for the Industrial Internet of Things, this paper reviewed existing literature. Specifically, this paper presents a review of cybersecurity testbeds, Industrial IoT systems' security goals, and challenges. To achieve this objective, the study randomly reviewed relevant literature and used snowballing techniques to identify relevant seminal works that were included in the paper.

The ensuing sections of this paper are as follows. Section 2 provides a background that touches on OT and IT integration and some attacks that have exploited Industrial systems. Section 3 provides an overview of cybersecurity testbeds, types of cybersecurity testbeds, cybersecurity testbed use cases and applications, and cybersecurity testbed challenges. Section 4 provides an understanding of the use case and application of the cybersecurity testbed, Section 5 highlights the challenges of the cybersecurity testbed, and Section 6 provides an overview of IoT and IIoT security challenges. Section 7 provides an overview of the Industrial IoT architecture framework. Section 8 provides an overview of the common Industrial Internet of Things protocols that are used in Industrial systems. Sections 9 provide an overview of the security goals of industrial systems, and sections 10 and 11 demonstrate the importance of virtual cybersecurity testbeds, respectively. Section 9 summarizes the findings from the review.

2 Background

ICS is designated as critical infrastructure because it is used for supervising and controlling industrial processes in real-time [4], [3]. According to [1], ICS depicts an interface where commands are initiated to control physical industrial processes. It encompasses both SCADA, representing the software-based element of ICS, and DCS, a process control system that connects field-layer devices such as sensors and actuators [1].

Industrial control systems are usually found in isolated industrial environments, but due to the need for efficiency, reliability, and productivity in modern industry processes, industries have embraced the integration of interconnected smart devices, sensors, and actuators for collecting and transmitting industrial data even across cloud platforms [4]. The fusion of operations technologies with information technologies is observed to form the basis of the Industrial Internet of Things (IIoT) [2], [5] highlight that IoT integration into industrial systems brings the possibility of automation and analytics, which is the cornerstone of cyber-physical systems. This makes it possible to monitor production systems and control physical aspects of industrial processes [5]. The benefits of this integration have led to the creation of services: smart factories, smart services, and smart products aiming at efficient processing, cost savings, and timely business decisions [5].

The handshake has proven to be beneficial in different sectors and industries. For example, the chemical sector is a complex value chain industry knitted together to form an interconnected system from plant operation, warehousing, and transportation to the storage system [6]. It provides the dam service providers with the ability to control and monitor dam facilities and systems remotely [7]. Likewise, in the defense sector, it brings efficiency and monitoring of complex systems from equipment design to production [8]. Similarly, the latest energy system depends on cyber-physical systems, thus requiring seamless interaction among connected components from production to distribution [9]. Agriculture in the area of smart farming and precision agriculture relies on smart and remote technologies [10]. The healthcare environment is full of a myriad of interactive medical devices for seamless healthcare delivery, real-time data collection, and data analytics [11]. The nuclear industry to safely monitor and control plants, and protect nuclear reactors, depends on interconnected industrial systems [12]. In the recent transportation system, the sector has increasingly depended on electronic systems for managing critical operations [13], [14], while the water and waste systems industry relies heavily on cyber-physical systems for water treatment and distribution [15].

This handshake (fusion of IT and OT) allows businesses to remain competitive by improving industrial processes, performing data analytics, making timely market-driven decisions, and sustaining production. However, it presents significant challenges, particularly in terms of security and privacy [2], [16], [5], [3], [17]. The interconnected nature of industrial settings makes cybersecurity a serious and frontal issue due to the presence of digital, analog, and mechanical components in an industrial environment [18]. This has resulted in increasing attack surfaces and vulnerabilities. Therefore, there is a need for holistic measures, encompassing the sensing layer, networking layer, middleware layer, and application layer [16]. Industrial IoT systems at various layers, such as communication and connectivity, are vulnerable to cyber-attacks [3]. Securing interconnected industrial systems requires hardware and encryption measures [5]. Ensuring the security of all tiers of IIoT architecture, from the edge layer to the enterprise layer, is crucial [3].

The same applies to the protection of vast amounts of data generated by Industrial systems, processed, and shared with different users and parties. Such sensitive information and its transmission from remote units to the cloud raise privacy concerns [19]. These concerns are related to social, legal, and technical domains. Hence, the need to protect the transmission of sensitive information from remote terminal units and storage systems, by complying with privacy regulations [2], [16], [3]. However, it is challenging to enforce privacy-related policies in an IoT environment [3].

According to [1], the gains for cyber actors are enormous when an important asset is successfully exploited or compromised. These potential gains have attracted diverse illicit cyber-related activities, not limited to cyber vandalism, cybercrime, cyber espionage, cyber terrorism, cyber sabotage, and cyber warfare. However, protecting industrial processes against illicit cyber activities is difficult and requires careful planning and extensive preparation. Protecting infrastructure from exploitation is challenging because it is difficult to pinpoint or know the intention of individuals engaging in malicious activities. This may be attributed to the global nature of cyberattacks and evolving organized crime groups targeting industrial digital assets.

Examples of these illicit cyber activities or cyber-attacks are Stuxnet, Hatman, Ragnar, PoisonIvy, and Shamoon. Stuxnet remains the most popular attack on Industrial digital systems, which specifically targeted SCADA systems [20], [8]. This zero-day attack exploits vulnerabilities in ICS aimed at sabotaging facilities, causing field-level devices to operate beyond their specified boundaries. This led to the failure of centrifuges at the Iranian nuclear plant [2]. The Stuxnet attack highlights the sophistication of well-funded adversaries targeting industrial equipment [20].

Other notable cyber-attacks on CI, as observed in [8], include the Ukraine power grid distribution attack in 2015, the Oldsmar water treatment attack in 2021, the Ragnar Locker attack from 2018 to 2020, and the German steel mill cyber-attack, which resulted in a furnace blast causing massive damage [1]. The Maroochy Shire Water Treatment facility attack in 2000 caused sewage spills and environmental pollution [4]. The PoisonIvy malware attack on the chemical factory website domain led to the theft of sensitive information from chemical factories in the US [1].

In 2017, the Hatman attack was launched against a Saudi Arabian petrochemical facility to disrupt Saudi Arabia's oil company supply chain. Similarly, in 2016, six Saudi Arabian organizations in the energy, aviation, and manufacturing sectors were targeted, and the firm's computers were wiped out. The LockerGoga ransomware attacks in 2019 were observed to disrupt Norway and US chemical manufacturing facility operations. Other cyber-attacks observed include two US national laboratories, resulting in a total shutdown of internet access and website operations, and the attempt of the Iranian cyber adversaries to release US Rye Brook Dam water in New York City [1]. Another sophisticated attack was the Chinese hacking malicious activity that led to the theft of technical documentation of the F-35 Joint Strike Fighter and the F-22 Raptor jet from the defense industrial base [1]. The Slammer worm in 2003 was used to disrupt The United States' nuclear power plants' monitoring systems, and in the same year, a US transportation network signal and dispatching control system was affected by a computer virus, resulting in freight services disruption and people movement impacted [2].

This noticeable shift and increase in cyber-attacks on critical infrastructure has attracted the interest of global actors such as governments, institutions, research, and the security community [21]. Efforts to address cyber-attacks include political, economic, organizational, legal, and technical measures to safeguard CI from threat actors [1], [22]. Of these stakeholders interested in understanding vulnerabilities and cyber-attacks on industrial settings are the research community and industry practitioners [3], [17]. However, practicing cyber-attacks on real-time systems limits and at the same time has significant implications when it comes to security goals, particularly the system availability [23]. Beyond the availability requirement, the inherent danger of conducting cyber-attack testing on production systems is significantly high in terms of monetary value [4]. Hence, the adoption of cybersecurity testbeds to closely prototype physical systems obtainable in the industrial environment for security testing and threat investigations [23].

Cybersecurity Testbed is a scientific instrument that empowers researchers and security professionals [23]. This allows the understanding and investigation of cyber incidents in the ever-growing cyber-physical systems. Although it faces integration and fidelity concerns [4], [23], cybersecurity testbeds remain and serve as alternate tools for the discovery of a vulnerability in an industry control system [23]. In addition, cybersecurity testbeds facilitate and bridge the barriers for new entrants and therefore make it easy for new researchers and industry professionals interested in acquiring first-hand cybersecurity knowledge of ICS operations [4]. The promising nature of cybersecurity testbeds and their flexibility has made them a cybersecurity tool for simulating different components of ICS infrastructure in various sectors such as energy, water waste management, transportation, and healthcare [24]. The use and relevance of cybersecurity testbeds continue to grow, although the concern of fidelity among others is of interest to the wider research and security community [23], [4].

In [23] scalability and interoperability are other challenges facing cybersecurity testbeds in terms of their use for different test scenarios. Despite this, cybersecurity testbeds are observed to be used for security testing, vulnerability discovery, and to demonstrate known attacks' impacts in industries such as Energy, water and sewage, aviation, health, oil and gas, telecommunication, agriculture, and defense.

According to [24], cybersecurity testbeds are essential because their applications for the security of the industrial infrastructure increased significantly from 2015 to 2023. This growth may be due to the relevance of cybersecurity testbeds and their utilization to leapfrog the cyber defense of ICS, and most especially, facilitate research.

3 Cybersecurity Testbeds

According to [25], IT and OT integration demands a new strategy to address new attack surfaces. This demand requires the gathering of quality and large datasets under different system conditions. It also calls for addressing data privacy and the need for a cost-effective system for designing test systems. This need paved the way for cybersecurity testbed development for research and education purposes. The possibilities can allow the research and security community to assess industrial infrastructure vulnerability and develop measures that aid the defense of industrial critical infrastructure.

Cybersecurity testbeds, according to [4], are platforms that allow for a variety of purposes: for a scaled-down version of real-world industrial control systems, to generate and observe behavior, and patterns of cyber-attacks. It also serves as a controlled environment where ICS vulnerabilities can be identified [4]. The environment helps researchers and practitioners visualize cyberattacks that are found in the most critical infrastructure. And they are used to develop mitigation strategies without risking actual operational environments, and for replicating part of the real industry for security practice [1]. Another characteristic of cybersecurity testbeds, according to [23], is their usage in preventing disruption during a security audit or testing practices. Studies such as [23], [26], [27], [15], [28] observed the application of cybersecurity testbeds for security testing, vulnerability discovery and analysis, threat analysis, testing defense measures, education and training purposes, simulation of industrial control systems, and performance analysis. In addition, the cybersecurity testbed makes it possible to observe attack propagation at different levels and stages of an industrial system in a controlled environment [8].

In complex cybersecurity compliance, where adhering to regulations, standards, and best practices to protect and defend industrial digital systems from threats is crucial, cybersecurity testbeds come in handy. It facilitates testing and validation of compliance and regulatory measures towards a risk-free environment. By simulating different compliance scenarios, organizations can build a robust cybersecurity posture and strengthen preparedness.

A similar mechanism that is observed from the literature is the use of a cyber range. The cyber range, according to [8] is a virtualized environment for training people to improve their response capacity. Cyber range depicts an environment being utilized by cybersecurity and non-cybersecurity experts to instantiate a cyber-attack exercise and their observations. While cybersecurity testbeds and cyber ranges are sometimes used interchangeably, they are different. The key difference is that cyber range is generalized; however, cyber security testbeds are specialized, focusing on specific systems to provide accurate fidelity for security research, testing, and validation [4], [8], [29], [30].

3.1 Types of Cybersecurity Testbeds

The uniqueness of cybersecurity testbeds and their importance in industry and research have attracted scholars' attention, resulting in a rich body of literature and a comprehensive review, resulting in the classification of cybersecurity testbeds. Cybersecurity testbeds are based on the design objective, the specificity of the research or operational objective, the nature of the components that are to be simulated, flexibility, cost, etc. Cybersecurity testbeds, as highlighted in [24] are classified into three primary categories: virtual, physical, and hybrid. Each type, however, has its advantages and disadvantages.

3.2 Virtual Cybersecurity Testbed

Virtual cybersecurity testbeds depict an environment equipped with virtualized elements where all components, including networks, field-level devices, and systems, are simulated using software and virtualization tools [24]. Although virtual environments and simulations may seem different [23], virtual tools enable the virtualization of actual ICS components and protocols rather than simulation. However, [25] considers both forms of virtualization. In addition, [8] highlighted that virtual testbeds use simulation, virtual machines, and emulation software to replicate a system. Some of these virtualization tools and software are mentioned in [24]

Virtual cybersecurity testbed is flexible and cost-effective, allowing rapid deployment and scalability [4], [23], [8]. However, they may not capture the full complexity of physical interactions and hardware-specific vulnerabilities. Often used in the context of developing scalable and flexible environments for cybersecurity research. For example, the use of virtual cybersecurity testbeds allows virtualizing an environment to simulate networks and threat events, enabling researchers to perform evaluations and validation without the constraints of physical hardware [29]. Such a platform provides a cost-effective solution for testing and cybersecurity strategies.

3.3 Physical Cybersecurity Testbed

Contrary to a virtualized environment, physical cybersecurity testbeds involve actual hardware and physical components, providing a realistic environment for physical system operations. However, they are often expensive to set up and costly to maintain [31], [15]. Moreover, operating real physical devices for experiments can sometimes limit research scope.

Physical cybersecurity testbeds provide high fidelity since they can closely reflect the environment where industrial systems intersect with cyber-systems [8]. In addition, it allows subjecting physical systems to actual security measures instead of simulated scenarios [32]. As a result, it helps to understand the behavior and condition of the systems. This is demonstrated by [15], [31] leveraging physical cybersecurity testbeds to defend water infrastructure to assess vulnerability and understand the propagation and cascade of attack impacts. These studies show the threat analysis of water distribution networks, and basically, how an attack on a cyber-physical system would affect another cyber-physical system. However, physical cybersecurity testbeds are prone to unauthorized manipulation and do not allow for collaboration, especially when it is not connected to the internet, particularly in terms of flexibility and reconfiguration [15].

3.4 Hybrid Cybersecurity Testbeds

Hybrid testbeds include elements of both virtual and physical testbeds. It reflects the blend of virtualization tools and some specific types of hardware. This type of testbed allows for comprehensive testing while balancing cost, complexity, and safety. The potential for cost-effectiveness, flexibility [29], and integration of both environments has increased its popularity and uses in the research community. It is an optimal solution for testing in an environment requiring both real-world components and simulation. For example, it described the PowerCyber hybrid testbed at Iowa State University for smart grid security testing by simulating both cyber-attacks and their physical impacts. Hybrid cybersecurity testbeds have seen increased deployment and use in industrial IoT [24]. This may be due to the hybrid nature of the cybersecurity testbed in the sense that it enables the possibility of testing scenarios [32].

4 Cybersecurity Testbed Use Case and Application

Understanding use cases of cybersecurity testbeds is essential for various reasons, especially when it comes to their application. It helps to know how cybersecurity testbeds are being applied in different sectors and their relevance in the current cybersecurity landscape. Figure 1 and Figure 2 illustrate how cybersecurity testbeds are used across different domains and use cases in the research community.

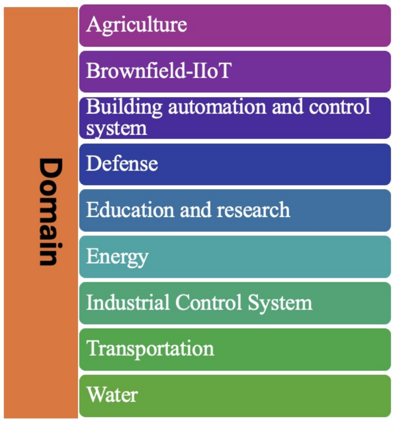

Fig. 1 cybersecurity application domain

In Agriculture, cybersecurity testbeds play a significant role in understanding IIoT systems in crop monitoring and automating irrigation systems. The cybersecurity testbed is used to evaluate and enhance security measures to protect agricultural technologies from cyberattacks [33], [10]. In Brownfield-IIoT, cybersecurity testbeds are used to understand how new IIoT technologies integrate with legacy systems. They allow for holistic testing and evaluation of ransomware attacks that target IIoT edge gateways, enabling researchers to analyze the impact and behavior of such attacks on brownfield deployments [34], [35]. Cybersecurity testbeds in building automation are used to assess the impact of cyberattacks on building management systems (BMS). For example, testbeds are deployed to simulate the security of HVAC systems, allowing researchers to conduct impact assessments and test defense mechanisms against potential cyber threats in building environments [36].

Fig. 2 cybersecurity testbed uses cases

The defense sector leverages cybersecurity testbeds to simulate the impact of cyber-attacks on defense-critical systems. A specific use case involves a physical testbed used to model a naval warship, allowing researchers to test cyber-attacks and develop countermeasures to protect industrial control systems in defense-critical environments [8]. Cybersecurity testbeds are widely used in the education and research industry because they enable researchers to study and understand cybersecurity threats. It serves as a platform for researchers to investigate vulnerabilities in various systems, test proposed solutions, and validate cybersecurity measures. The flexibility of virtual and hybrid testbeds makes them particularly valuable for cost-effective research and scalability in academic settings [4], [37], [38], [39], [40].

The energy sector heavily utilizes cybersecurity testbeds to protect critical infrastructure, including distribution systems, generation facilities, substations, and renewable energy sources. However, they are largely used for researching, observing, and testing security measures to protect against cyber-attacks that could disrupt the energy supply [41], [42], [43], [44], [45]. Another application area of Cybersecurity testbeds is used to simulate industrial control systems (ICS) and to evaluate and secure critical infrastructure. The simulations help researchers understand cybersecurity threats and assess the resilience of ICS, enabling them to implement appropriate security measures to protect against potential attacks [46], [47], [48], [49]. The transportation industry leverages cybersecurity testbeds to test security solutions related to traffic management systems, in-vehicle networks, and vehicle communication systems. An example includes the use of testbeds to investigate the security of automotive CAN bus systems and assess the effectiveness of security controls in preventing unauthorized vehicle control [14], [13]. In addition, cybersecurity testbeds are used in the water and waste industry to assess the security of water treatment and distribution systems. This mechanism helps detect anomalies in water distribution processes and protect critical water infrastructure from cyber-attacks [15], [31], [50], [51]

Figure 2 illustrates the uses of cybersecurity testbeds. They are mainly used for cyber-attack assessment and analysis, threat detection and mitigation, forensic analysis, security testing, privacy and data security, education and research, and ICS simulation[52], [24]. Cybersecurity testbeds support and enhance knowledge of cybersecurity issues in IIoT by analyzing and assessing vulnerabilities in a controlled environment. They facilitate operational continuity and test data integrity. The environment is equipped with tools like Wireshark and Metasploit for the simulation of various cyber-attack scenarios, helping to defend and improve defense capabilities. Cybersecurity testbeds are likewise used for forensics analysis, allowing investigators to trace the origin of attacks and understand the tactics used by cybercriminals [53].

In addition, cybersecurity testbeds facilitate the resilience of IIoT systems through security testing and validation, particularly in federated interconnected systems. In addition, cybersecurity testbeds help to understand data security and privacy challenges in IIoT systems, while ensuring cybersecurity compliance with various regulations such as privacy regulations.

5 Cybersecurity Testbed Challenges

Despite the many benefits of cybersecurity testbeds, it is not without challenges in terms of design and use. According to [23], developing cybersecurity testbeds that scale and achieve a realistic configuration is a difficult task. This could be attributed to the levels of rigor and skills required to design and configure cybersecurity testbeds that closely adapt to real systems. While cybersecurity testbeds are pronounced in the academic environment, designs that aid collaboration and coordination among researchers remain. The borderless nature of cyberattacks requires collaboration and coordination. Another challenge with cybersecurity testbed applications is ensuring that the data generated reflects what is obtainable in the real environment, hence the concern of fidelity.

Most studies that utilized cybersecurity testbeds did not describe how their design meets fidelity requirements or provide enough metrics for replication [23]. The absence of such requirements and metrics raises validity and reliability

Fig. 3 Cybersecurity testbed challenges

concerns. In addition, there is no holistic framework for comparing the fidelity of cybersecurity testbeds, which further complicates the issue. However, [54] outlined four criteria that cybersecurity testbeds should satisfy, but [23] argued that two of these criteria, repeatability and measurement accuracy, go beyond the technicality of testbed design. As shown in Figure 3, cybersecurity challenges, in addition to other challenges, include cost and data collection and sharing.

1. Fidelity: Ensuring that cybersecurity testbeds closely reflect real-world conditions, such as mimicking relevant physical processes and environmental factors, is challenging but important when replicating complex industrial safety systems and critical infrastructures [23], [54]. Designing cybersecurity testbeds that closely reflect a real-world Industrial Control System (ICS) involves a combination of factors such as physical processes, network architecture, and integration of the control system. The design requires specialized domain expertise and clear design guidelines, which are often not readily available, particularly when attempting to replicate real-world industrial scenarios [23]
2. Repeatability: The lack of documentation makes it difficult for other researchers to replicate experiments or understand the cybersecurity testbed's setup and configuration [54],

[23]. The challenge of repeatability may occur when systems (proprietary or open-source systems) and procedures that were used are poorly documented.

3. Measurement of accuracy: The inaccurate representation of system behavior would affect the outcome, hence the challenge of measuring accuracy and general concern about the fidelity of cybersecurity testbed [54]
4. Interference or safe execution: The interference or safe execution criteria target safe execution whereby tests conducted with cybersecurity testbeds do not affect the performance or negatively affect the system [54].
5. Cost: Building and maintaining physical testbeds, which provide the highest fidelity, are expensive and resource-intensive. This includes the cost of procuring industrial hardware, configuring the environment, and ongoing maintenance. Finding the necessary funding resources for long-term testbed projects is often difficult. This may be why most cybersecurity testbeds are funded projects mainly from governmental institutions [24]
6. Safety: Running experiments on physical testbeds that control real industrial processes can be risky. Attacks simulated on these testbeds can potentially cause physical damage to the equipment or even endanger human lives, making it difficult to test certain scenarios safely [23]. Safety concerns can limit and hinder security testing and exercises because of the fear of damage to the physical system. Scaling cybersecurity testbeds presents a significant challenge, especially when diverse systems are integrated or complex scenarios are simulated. Cybersecurity testbeds, especially physical ones are resource-constrained. They are limited by space and location, hence the problem of expansion [15], [31].
7. Data Collection and Sharing: Collecting and sharing high-quality data from testbeds is often difficult. Challenges include capturing relevant network traffic, processing data accurately, and labeling datasets for use in machine learning or other analytical methods [55].

6 IoT and IIoT Security Challenges

Integration of the IoT in an industrial setting exposes industrial systems to lots of challenges. Some of these challenges are present in IoT and IIoT environments, while some are unique to the industrial internet of things. As highlighted in [17], challenges in both environments are related to data confidentiality, integrity, key establishment of device pairing, device management, and ICS system security. Data Confidentiality: Data confidentiality is a top concern in both IoT and IIoT due to the sensitive nature of data generated and processed [2].

IIoT systems, which are often found in critical industries, produce vast amounts of raw and non-uniform data that require real-time processing and secure transmission. Securing data transmission requires encryption. However, the challenge lies in implementing a scalable encryption mechanism that can be applied within the constraints of IIoT devices. The computational resource limitation of IIoT systems makes it more challenging to develop efficient encryption algorithms. The risk of data breaches in IIoT systems is high and of great consequences, hence the necessity for advanced encryption techniques capable of securing IIoT data at scale.

1. Integrity: Integrity of industrial IoT is a critical security concern in both IoT and IIoT environments. Because they are responsible for monitoring and controlling physical processes, they have become a prime target for malicious attacks. Ensuring the integrity of these systems involves verifying the software and hardware configurations through methods such as attestation [2]). However, existing attestation techniques often rely on assumptions and computational resources that may not be feasible for low-cost and resource-constrained IIoT devices. Developing practical and secure attestation mechanisms that can effectively verify the integrity of IIoT systems without imposing excessive computational overhead is required.
2. Key Establishment for Device Pairing: Secure device pairing and key establishment are important for ensuring secure communication in IIoT systems. This challenge is particularly in IoT environments where devices are often resource-constrained and produced by various manufacturers, making use of pre-loaded shared secrets impractical. Industrial IoT systems require robust key establishment protocols that can securely pair devices and facilitate cryptographic communication without human intervention. While context-based pairing solutions have been proposed, they often rely on specific sensing capabilities that may not be universally available, posing a significant challenge in heterogeneous IIoT environments.
3. Device Management: Managing and securing is a critical Industrial IoT, given the scale and complexity of industrial deployments [2] is challenging. Device management is the governance and tracking of devices throughout their lifecycle. It includes monitoring status, firmware, and patch updates while managing relationships between devices. In the context of IIoT systems, the challenge is more pronounced because of the need to manage large numbers of devices in real-time.

In addition, the Industrial Internet of Things has specific security challenges because of the integration of IT infrastructure and traditional industrial systems [17]. Traditionally, ICS systems were isolated and were not exposed to digital threats. However, today's interconnected nature of industrial systems has increased exposure and ICS susceptibility to cyberattacks [23].

Protecting IoT gateways within IIoT architecture faces significant cyber threats because of their roles in bridging field devices and service clouds. And how they facilitate access to silo data from sensors and embedded controllers [17]. This device makes it possible to carry out remote preventive maintenance and ensure process optimization. IIoT gateway's critical roles make it a prime target for threat actors. Hence, the need for a secure IoT gateway for various industrial applications [2].

Harnessing the potential of the integration of industrial robots within the IIoT environment has increased the attack surface. These highly automated robots bring efficiency to manufacturing in terms of assembling, pack- aging, assembling, painting, and product inspection [30]. Because they are connected to external enterprise systems and cloud services via communication protocols introduce new attack surfaces. Understanding

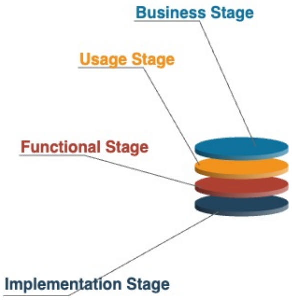

Fig. 4 IIoT architecture framework

the vulnerabilities in such systems and their defense would require the deployment of tailored cybersecurity measures.

7 IIoT Architecture Framework

Two important frameworks for Industrial Internet of Things are observed in the literature. These frameworks served as reference guidelines for understanding layers of cyber-physical systems and their interaction. Following this reference architecture guides the design of cybersecurity testbeds that include the necessary components and stages.

1. Industrial Internet Consortium (IIC) Reference Architecture: IIC reference architecture provides a detailed framework for understanding the interaction of cyber-physical technologies in IIoT systems [56]. It breaks down the architecture into multiple stages as reflected in Figure 2. This includes business, usage, functional, and implementation viewpoints or stages.

   The business viewpoint addresses the business need and objective of establishing Industrial IoT systems. The usage point depicts how the system will be used, including the expected system usage. The functional point depicts the functional component, system structure, and their interactions, while the implementation represents the architecture and technical description of the Industrial IoT system. The IIC framework emphasizes security across all tiers, including support for authentication, cryptographic protection, secure key storage, and interoperability across multi-vendor systems [3]
2. OpenFog Consortium Architecture: The OpenFog Consortium's architecture focuses on bringing processing capabilities closer to the edge of the network through fog computing [3]. The design complements the use of cloud technology to address performance and efficiency, resulting in enhancing service performance [57], which makes it relevant and can be applied in industries such as smart cities, finance, healthcare, agriculture, transportation, etc.

   This framework consists of three layers: communication, service, and application security. It also introduces two operational planes focusing on security provisioning, monitoring, and management, which are designed to handle the unique security and processing requirements of fog-based IIoT systems [3], [2]. The communication layer takes care of data transmission between fog nodes, devices, and the cloud. The service layer handles service coordination and accessibility to computing resources. The application layer is concerned with securing applications running in the fog environment. To ensure the security of devices and data in fog computing, there is a need for holistic security mechanisms.
3. Another framework highlighted in [4] is the Purdue model, widely used in

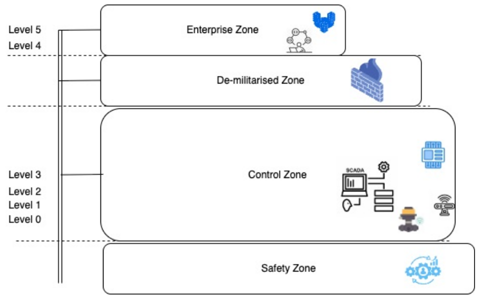

Fig. 5 Purdue model framework

the design of an industrial control system, especially in manufacturing or the design of critical systems. As illustrated in Figure 3, the framework segments the industrial system into zone layers and levels, with each representing specific functions and operations. The work in [8] guided by this framework was able to accurately describe their testbed components and the level, respectively.

According to [4], the Purdue framework facilitates a defense-in-depth strategy for securing the ICS system. The framework is divided into 4 zones and 6 levels, respectively. The Enterprise zone comprises corporate IT infrastructure and a business-logic environment. Enterprise environment, which is level 5, consists of IT infrastructure connected to the internet and used by the company staff. Level 4 is the business logic consisting of the traditional portion of the network, while the Demilitarized Zone (DMZ) separates and acts as a buffer between the Enterprise Zone (IT) and the Operational Technology (OT) in the Control Zone. The objective is to facilitate secure data exchange while protecting OT systems from unknown threats.

The Control Zone consists of levels 0 to 3. Level 3 is the operations management responsible for overseeing production activities. Examples of tasks at this level are scheduling and quality control. Level 2 is the supervisory level, which comprises SCADA and HMI systems for monitoring and controlling industrial processes. Level 1 is where real-time control operations occur. It manages the PLCs, and other intelligent devices, while the process level consists of actual physical processes, consisting of sensors, actuators, and machines that interact with the physical world. The safety zone is an independent zone that includes safety instrumented systems purposely designed to ensure ICS operates in a safe state. It is designed to intervene when observed that ICS operates in unsafe conditions.

8 Common industrial Internet of Things Protocols

Several protocols, particularly for the industrial IoT are observed in the literature [58], [4], [3], [26]. Some of these protocols are related to communication and security protocols [18].

1. Modbus/TCP: Modbus/TCP is a known communication protocol widely adopted in industrial automation and control systems for connecting electronic devices. It is used for communicating with industrial devices, such as connecting PLCs and sensors to the industrial network while facilitating data acquisition and process control [4], [8].
2. MQTT: MQTT is a lightweight publish/subscribe protocol commonly used in Industrial IoT, but it lacks built-in security guarantees.

To address this, SMQTT introduces lightweight encryption mechanisms, making it more suitable for secure data transmission in resource-constrained environments [3].

3. DNP3: DNP is a distributed network protocol mostly used in the utilities sector, such as water and energy, to facilitate communication between control systems and SCADA (Supervisory Control and Data Acquisition) systems [8], [29].
4. Zigbee: Zigbee provides secure communication at the network layer with AES encryption and node authentication. It supports low-power wireless communications and is commonly used in IIoT environments due to its robustness and energy efficiency [18].
5. LoRaWAN: This protocol offers long-range, low-power communication with AES encryption for data security. However, certain vulnerabilities like susceptibility to jamming attacks and weaknesses in cryptographic primitives have been identified, necessitating ongoing improvements.
6. BACnet: The BACnet is a communication protocol designed for building automation and industrial control systems. Among the several communication options, BACnet master-slave is the most widely used in industrial systems. Because the protocol lacks a data confidentiality mechanism, attacks such as reconnaissance and Denial of Service are possible [26]
7. Profibus: This communication protocol is widely used in industrial automation systems, allowing data exchange between field devices and controller systems, especially for linking and connecting devices from different vendors [58]
8. OPC: OPC is an open platform for communication. It is a communication protocol used in a client-server architecture for sending and receiving information in an industrial automation environment [59]. However, this standard has been improved (OPC UA) to ensure reliable communication and defend against cyber-attacks [60].

Fig. 6 Industrial Internet of Things security goals

9 Security Goals of Industrial Systems

Industrial systems require different security measures to assure confidence and trust. Figure 6 highlights the security goals when considering holistic cybersecurity measures to address the challenge of security and privacy issues in Industrial IoT. These goals include authentication, access control, confidentiality, identity management, availability, integrity, trust, safety, accountability, and encryption. Achieving the security goals of an industrial IoT will require a combination of security protocols, hardware components, and other measures [3].

Implementing an access control mechanism and the use of cryptographic techniques to enforce access rights is essential to prevent unauthorized access to sensitive data and systems [3]. The use of encryption across various communication layers will ensure that data remains confidential from level 0 to the destination.

Achieving this requires both software and hardware encryption methods.

For an organization to protect intellectual property, only authorized parties are required to access and use specific data or processes. This process will require a myriad of security measures and enforcement of policies to secure confidential information within the industrial systems [2].

Securing industry infrastructure is continuous, thus requiring continuous evolving monitoring and updating of security measures. An understanding of Industrial IoT security goals would facilitate the necessary actions to keep the system safe from malicious activities. Establishing and maintaining device identity and services provision requires authentication and authorization to enable entities to seamlessly interact [2]

The use of cybersecurity testbeds, therefore, allows researchers and practitioners to assess and investigate cyber threats that may compromise the security goals of Industrial IoT systems. Conducting experiments with known attacks helps in understanding the specific impact on each security goal as it manifests within the system.

10 Virtual Cybersecurity Testbed

The adoption of virtual cybersecurity testbeds and their relevance in research are important due to their cost-effectiveness and flexibility, particularly in terms of scalability, as well as their suitability in adapting to the dynamic cyber threat landscape. This could be the reason why virtual cybersecurity testbeds are one of the environments for security testing and assessment in the context of the Industrial Internet of Things.

Since cyber-attacks and cyber threats have no boundaries or limitations, organizations operating in an industrial environment, such as those engaged in industrial equipment manufacturing or production, require an alternative environment to test realistic security measures and detect anomalies in operation. This understanding is reflected in the work of [8], where cybersecurity testbeds were deployed to design a realistic environment and to investigate cyber-attacks. Another is the growing concern of attacks on industrial systems, which have resulted in stringent regulations leaving industrial organizations with compliance responsibilities. Complying with standards and regulations in the fast-growing Industrial IoT requires organizations to be proactive in building a robust cybersecurity environment and culture. This requires a proactive approach to identify, assess, and mitigate risk to critical infrastructure such as SCADA, HMI, PLC, sensors, actuators, etc. For instance, [4] leveraged a virtual cybersecurity testbed beyond security testing to evaluate risk in critical industrial systems based on cybersecurity standards [4]. This paper aligns with the opinion of this study on the potential of virtual cybersecurity testbeds to help organizations mitigate risks to critical infrastructure while meeting compliance requirements.

To prepare and respond to cyber threats, organizations can leverage virtual cybersecurity testbeds to assess and mitigate risk to Industrial IoT systems through the simulation of compliance requirements in the virtual environment, resulting in an improved security posture of industrial systems. Virtual cybersecurity testbeds can help industrial practitioners monitor critical industrial systems while adhering to safety and cybersecurity regulations. Researchers can also use the environment to test and evaluate security measures against attacks, protocols, and standards for improving the security of industrial systems. This could be why the security community and researchers adopt cybersecurity testbeds as highlighted in [24], [25], however, hybrid cybersecurity testbed usage is high when there is a need to balance fidelity requirements. As the software industry evolves, it will enable the possibility of simulating more industrial components and protocols. This in turn, drives the continued use and relevance of virtual cybersecurity testbeds in the field of cybersecurity.

Cybersecurity testbed design requires thorough planning to meet the requirements as highlighted in [23], [61]. To help facilitate the planning process, the study highlights key steps that may be followed when designing virtual cybersecurity testbeds. These steps include defining the objective of the cybersecurity testbed, scope, regulatory requirements, architecture framework, security goals, threat scenarios, scalability, collaboration, data collection, and validation. Defining clear objectives is essential for developing a successful virtual cybersecurity testbed as well as the scope. It is impossible to have a generic environment that applies to all systems, hence the need for scoping. In the industrial environment, data processing and data security are major concerns. Therefore, it is important to consider appropriate regulations while designing a cybersecurity testbed.

The design of cybersecurity testbeds needs to follow certain architectural frameworks. Following a standardized architectural framework in cybersecurity testbed design allows for flexibility in testing different ICS or IIoT components and determining appropriate security measures. Security goals of IIoT systems are another factor that needs to be considered when developing a virtual cybersecurity testbed. Having a clear understanding of the security goal to be tested will enable the simulation of specific attacks and components within the IIoT systems. Another is the attack scenarios. Successful attack scenarios in a controlled environment need to be planned and documented to aid the understanding of the attack behavior as well as the target system.

A well-designed virtual cybersecurity testbed would allow cross-collaboration among researchers and practitioners. The same applies to scalability when the need arises to reconfigure to accommodate new devices and protocols. The ability to quickly redesign will allow for rapid iteration of security testing as new security challenges are discovered while facilitating continuous research.

Often, data collection and analysis are not thoroughly planned when developing cybersecurity testbeds. To be able to understand attack behavior and its exploitation, selecting the right data collection and analysis tools is essential for collecting attack behaviors and system raw data. Documenting this process will facilitate quick replication and validation, making it especially useful during upgrades and maintenance.

11 Summary

Cybersecurity testbeds among the cybersecurity and research community have become a useful tool to facilitate security testing, research, vulnerability assessment, threat detection, and cybersecurity compliance. It aids in improving the security posture of the industrial system while adhering to standards and regulations.

This paper offers insight into the importance and urgency of protecting critical infrastructure, especially industrial IoT. It highlights the relevance of cybersecurity testbeds and their applications in different sectors such as Water and Waste Management, Energy, Dam, Agriculture, Defense, Transportation, Education and Research, and Simulation of Industrial Control Systems. The types of cybersecurity testbeds (physical, virtual, hybrid) observed are based on several factors: objective, flexibility, systems to be simulated, cost, and time, respectively. Each type of cybersecurity testbed, therefore, has its own advantages and disadvantages.

Since the growing attacks and cyber threats have no boundaries or limitations, organizations with an industrial environment need flexible and adaptable tools such as virtual cybersecurity testbeds to test security measures and to detect anomalies in operation. A virtual cybersecurity testbed, which serves as a controlled environment, will enable the simulation of known attacks and reconfiguration to suit different purposes with minimal resources. Such possibilities, as well as the growth of the software industry, will continue to make virtual cybersecurity testbeds relevant in the field of cybersecurity. Therefore, a virtual cybersecurity testbed is essential to keep pace with the complexity of the cybersecurity threats and compliance landscape while helping organizations to improve their security posture and devise security strategies to mitigate risk to industrial systems.

To minimize the challenges associated with designing effective virtual cybersecurity testbeds, research, and the cybersecurity community would need to ensure thorough planning and follow specific frameworks or guidelines that will ensure virtual cybersecurity testbeds closely mimic the real-world environment, and can therefore be reproducible and validated.

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